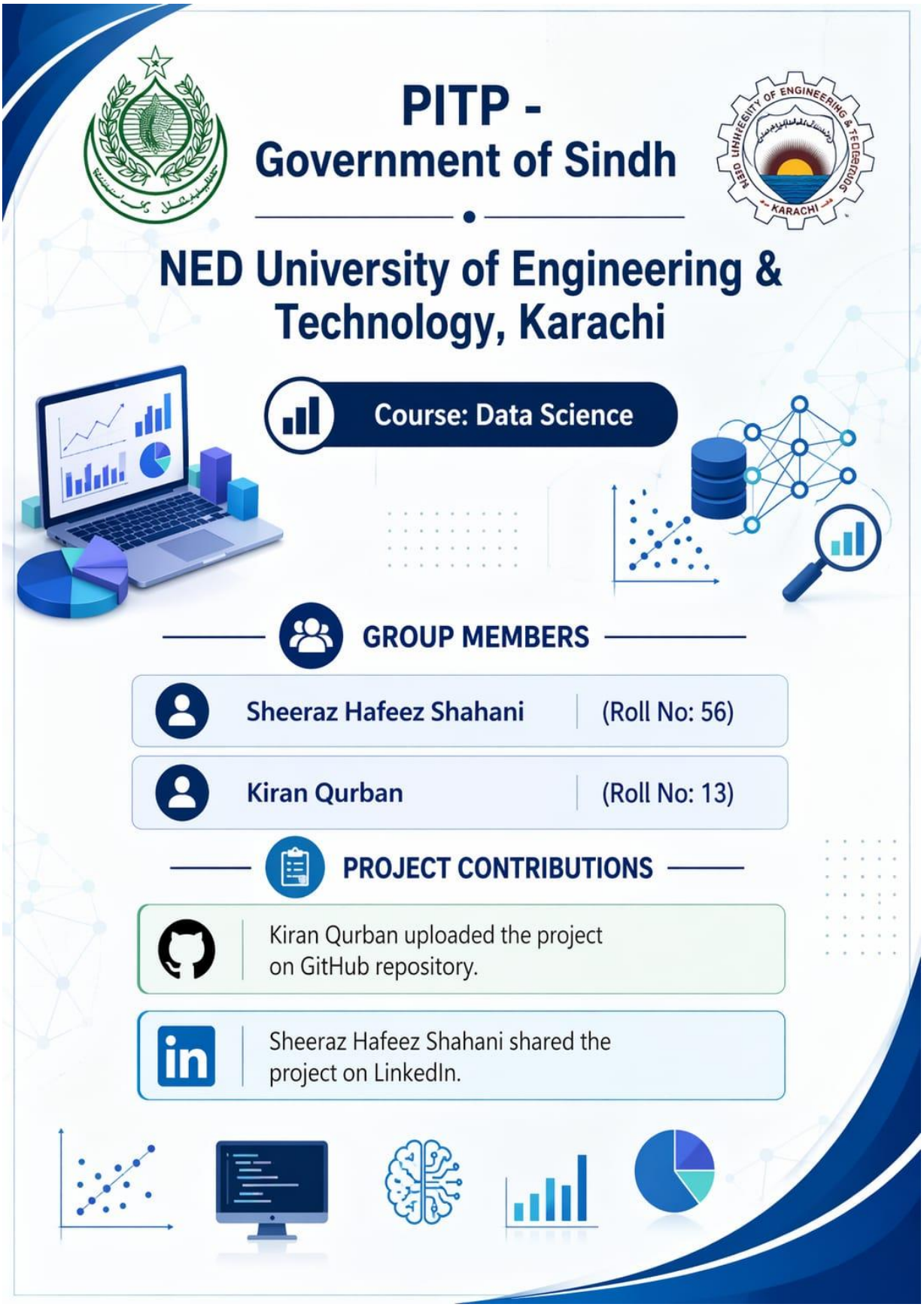

Report Diabetes Prediction Using Machine Learning

Introduction

This project is based on the Pima Indians Diabetes Dataset obtained from Kaggle. The dataset contains medical records used to predict diabetes occurrence in patients. It includes features like glucose level, BMI, age, insulin, and blood pressure. The main objective is to build a machine learning model for early diabetes detection.

Dataset Description

The dataset is taken from Kaggle: Pima Indians Diabetes Dataset. It contains 768 samples with 8 input features and 1 output label. The output indicates whether a patient has diabetes or not. It is widely used for healthcare machine learning projects.

Data Preprocessing

Missing values were handled by replacing zero values with mean/median. Outliers were detected using boxplots and removed using IQR method. Data was standardized using StandardScaler for better model performance. Finally, dataset was split into training and testing sets (80/20).

Exploratory Data Analysis (EDA)

EDA was performed to understand data distribution and relationships. Histograms were used to visualize glucose, BMI, and age distribution. Correlation heatmap showed strong relation of glucose with diabetes. Scatter plots helped in identifying feature patterns.

Histogram: Shows distribution of glucose, BMI and age. Some features are skewed.

Boxplot: Used to detect outliers in BMI, insulin and glucose values.

Correlation Heatmap: Shows strong relationship between glucose level and diabetes outcome.

Scatter Plot: Shows that higher glucose levels increase the chance of diabetes.

Feature Engineering

A new feature BMI category was created for better classification. Feature selection was performed using correlation analysis. Irrelevant features were reduced to improve accuracy. This step improved overall model performance.

Model Building

Three machine learning models were used for prediction. These include Logistic Regression, Decision Tree, and Random Forest. Models were trained on processed training data. Each model was tested on unseen test data.

Model Evaluation

Models were evaluated using accuracy, precision, recall, and F1-score. Confusion matrix was used to analyze predictions. Random Forest showed the highest performance. Results were compared to select the best model.
Model Evaluation and Accuracy:

Decision Tree Accuracy: 72.72% (0.7272)
Logistic Regression Accuracy: 75.30% (0.753)
Random Forest Accuracy: 75.30% (0.753)

Random Forest and Logistic Regression performed best due to higher accuracy and stable results.

Results

Random Forest achieved the best accuracy among all models. Glucose level was identified as the most important feature. Model showed good balance between precision and recall. Final model was selected based on performance metrics.

Conclusion

The project successfully predicts diabetes using machine learning techniques. It demonstrates the importance of preprocessing and feature selection. Healthcare data can be effectively used for early diagnosis. The model helps in improving medical decision-making.

Future Scope

The project can be improved using deep learning techniques. Larger datasets can increase accuracy. Real-time deployment can be done using web applications. It can also be extended to other diseases prediction.

Tools Used

Python programming language was used for implementation. Libraries like Pandas, NumPy, and Scikit-learn were used. Matplotlib and Seaborn were used for visualization. Google Colab was used for code execution.

GitHub & LinkedIn Activity

The complete project code is uploaded on GitHub. It includes dataset, code, and trained models. The project was also shared on LinkedIn for professional exposure. It highlights skills in Data Science and Machine Learning.

Google colab notebook link

https://colab.research.google.com/drive/1a5G6ps2M6Z-uMooQbnC_FvJSYtKnKMlB?usp=sharing

THANK YOU